Course Code & Name: CSA1510 Cloud computing ASSESSMENT Tool : CO2_AT1	Name: K. Saipriya RegNo: 192421296

Q1. Compare Type 1 and Type 2 hypervisors for a cloud data center hosting mission-critical applications. Analyze performance, security, manageability, then justify the better choice.

COMPARISON TABLE

Feature	Type-1 Hypervisor (Bare Metal)
Installation	Runs directly on physical hardware
Performance	Very high performance with low overhead
Security	More secure because there is no host OS attack surface.
Manageability	Centralized management, easy to monitor and scale
Reliability	High reliable with minimal downtime.
Examples	VMware ESXi, Microsoft Hyper-V, Xen

Analysis :

Performance: Type -1 hypervisors run directly on the hardware, so they provide faster processing, lower latency, and better resource utilization. Type -2 hypervisors depend on the host operating system, which adds extra overhead and reduces performance.

Security: Type-1 hypervisors are more secure because they have a smaller attack surface and do not rely on a host operating system. Type-2 hypervisors are more vulnerable since an attack on the host OS can affect all virtual machines.

Manageability: Type-1 hypervisors support centralized management, live migration, load balancing, and high availability, making them suitable for large cloud datacenters. Type-2 hypervisors are mainly used for testing, development, and personal use.

Justification:

for Mission - critical cloud applications, Type-1 HyPervisor is better choice because it Provides:

* High Performance with low overhead
* Strong Security through hardware - level Virtualization.
* Better reliability and high availability
* Easy centralized management and Scalability for enterprise cloud environments.

Conclusion:

Type-1 HyPervisor is the most Suitable choice for a cloud datacenter hosting mission-critical applications because it offers Superior performance, Stronger Security, and better manageability than a Type-2 Hypervisor.

Q2. A cloud provider must isolate workloads belonging to finance, healthcare and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Analysis Table:

Aspect	Explanation
Virtualization Architecture	A hypervisor creates multiple Virtual Machines (VMs) on the same physical server. Each VM has its own CPU, memory, storage, operating system ensuring workloads remain isolated
Finance workload	Runs in a dedicated VM with strict access control and encrypted storage to protect financial data.
Health care workload	Runs in a separate VM to secure patient records and maintain privacy.
Student portal workload	Operates in another isolated VM, preventing interference with finance and health care applications.

How Virtualization Supports Isolation

* Each workload returns in a separate virtual Machine (VM)

* The hypervisor isolates cpu, memory, Storage, and network resources between VMs.

* If one VM is compromised, the other VMs continue to operate securely.

* Resource allocation and access control prevent unauthorized access between workloads.

Two Major Attack Surfaces

Attack Surface	Description	Prevention
Hypervisor Attack	Attackers exploit vulnerabilities in the hypervisor to gain control over multiple VMs.	Regular patching, secure configuration, and access control
VM Escape Attack	An attacker breaks out of one VM and gain access to the host or other VMs	Strong VM isolation, updated hypervisor, and Security monitoring

Conclusion:

Virtualization architecture enables multiple work loads to run safely on the same hardware by providing strong isolation between virtual machines. However, Hypervisor Attacks and VM Escape Attacks are major security risks that can be minimized through proper security measures, updates, and strict access control.

Q3. Study the following VM host utilization snapshot and recommend corrective actions : CPU 92%, Memory 88%. Storage I/O latency 35ms, Average VM boot time 170s Explain the likely bottlenecks.

Analysis Table:

Parameter	Current Value	Likely Bottleneck	Corrective Action
CPU utilization	92%	CPU is overload, causing slow VM Performance	Add more CPU resour balance workloa

Memory utilization	88%	High memory usage may lead to swapping and reduced performance	Increases RAM, optimize memory allocation
Storage I/O latency	35ms	Slow disk access delays data read/ write operations	upgrade to SSD/NVMe storage, distribute storage load
Average VM Boot Time	170s	Slow boot due to cpu, memory and storage bottlenecks	Improve cpu, memory, and storage performance to reduce boot time

Analysis

* CPU Bottlenecks: cpu utilization of 92% indicates the host is heavily loaded, reducing VM performance.

* Memory Bottleneck: Memory utilization of 88%. Suggest limited available RAM, which can cause slow response times

* Storage Bottleneck: Storage I/O latency of 35Ms is high, resulting in slower disk operation

* VM Boot Time: An average boot time of 170 s shows that system resources are insufficient and storage performance is affecting startup.

Recommended Corrective Actions

* Add more CPU cores or migrate some VMs to another host.

* Increase Physical Memory and optimize VM Memory allocation.

* upgrade Storage to SSD / NVMe and balance Storage workloads.

* continuously monitor resource utilization and enable load balancing to avoid resource saturation

Conclusion:

The VM host has high CPU, Memory and Storage usage, causing performance issues. upgrading resources and balancing workloads will improve VM performance and reduce boot time.

Q4. How does Software Defined Datacenter (SDDC) improve agility compared to a traditional datacenter, Explain using compute, Storage & Network virtualization

Comparison Table:

Aspect	Traditional Datacenters	SDDC
Compute	Physical Servers	Virtual Machines
Storage	Hardware - based	Virtualized Storage
Network	Manual configuration	Software - defined networking (SDN)
Agility	Slow deployment	Fast and Scalable

Analysis:

* Compute Virtualization:

Multiple virtual Machines can run on one physical Server, improving resource utilization and enabling quick VM provisioning.

* Storage Virtualization:

combines storage resources into a single virtual pool, making storage easier to allocate, expand, and Manage.

* Network Virtualization:

Uses Software-Defined Networking (SDN) to centrally manage network configurations, improving flexibility and reducing manual effort.

Advantages of SDDC:

* faster deployment of applications
* Better Scalability and flexibility
* Automated resource Management
* Reduced operational cost
* Improved resource utilization.

Conclusion:

SDDC is more agile than a traditional datacenter because it automates compute, Storage, network Management, enabling faster deployment and better scalability.

Q5. Analyze the issue and propose a secure identity and privacy desing using federation concepts for multi-cloud federation project that fails due to identify mismatches between providers.

Analysis Table:

Issue	Solution
Identify mismatch between cloud providers	use federated identity Management (FIM)
Different login credentials	Implement Single Sign-on (SSO)
unauthorized access	Apply multi-factor Authenticator (MFA) and Role-Based Access control (RBAC).
Privacy risks	Encrypt user data and use Secure authentication protocols

Analysis:

* Identity mismatches occur because each cloud provider uses different authentication methods.

* Federated Identity Management (FIM) allows user to access multiple cloud services with a single identity.

* Single Sign-on (SSO) reduces the need for multiple usernames and passwords.

* MFA and RBAC improve security by verifying users and limiting access based on roles.

* Data encryption protects user privacy during storage and transmission.

Conclusion:

Using FIM, SSO, MFA, RBAC and encryption provides secure authentication, protects user privacy, and enables seamless access accross multiple cloud providers.